Multimodal Medical Image Parsing with Task-Specific Low-Rank Adapters

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Abstract. Adapting a single vision-language model to medical image parsing requires training choices that reflect the demands of each task. We study task-specific adaptation with Qwen3.5-9B for FLARE 2026 Task 3, covering six task families and seven imaging modalities. Our system couples a shared, frozen 4-bit backbone with LoRA adapters selected from the input question. We customize adapter ranks and prompts, rebalance rare classification targets, and supplement regression training with dermatology classification examples. On the fixed internal holdout, the task-specific adapters achieve a dataset-macro balanced accuracy of 0.7791 for classification, a dataset-macro micro-F1 of 0.6286 for multi-label classification, a counting mean absolute error (MAE) of 11.73, a dataset-macro detection F1 of 0.8914, a dataset-macro regression MAE of 9.78, and a report-generation CRIMSON score of 0.8204. On the organizer's 1,703-example hidden-validation set, the submitted system achieves 0.8496 classification balanced accuracy, 0.5338 sample-averaged multi-label F1, and 14.5950 regression MAE. The final 30.1 GB container averages 3.1 seconds per case, including model loading, and produces identical outputs across repeated verification runs. The benefits vary across tasks, supporting the use of task-specific adapter configurations and image budgets within a shared medical vision-language model. Code is available at https://github.com/TahaSabir0/FLARE26-Hexapus.

Keywords: Medical image parsing · Vision-language models · Parameter-efficient fine-tuning · LoRA · Task routing · FLARE 2026.

1 Introduction

Unlike conventional visual question answering (VQA), where the expected response is primarily textual, medical image parsing requires a general medical vision-language model (VLM) to interpret heterogeneous visual inputs and produce task-appropriate outputs ranging from categorical and multi-label predictions to numerical estimates, bounding boxes, and free-form reports, across modalities with different visual characteristics and anatomical morphologies. Medical VLMs range from biomedical dialogue systems such as LLaVA-Med [10] to models pretrained across clinical imaging domains such as MedGemma [16], and benchmarks compare generalist and specialist performance across clinical domains [24]. Med-MoE [9] illustrates the need to combine discriminative and generative medical tasks, while a systematic task-transfer study [15] shows that fine-tuning one visual capability can unpredictably improve or degrade others.

FLARE 2026 Task 3 [6] provides a practical setting for studying this problem. After data cleaning, its 51,154 question-answer pairs and 50,996 images span 19 datasets, eight imaging modalities, and seven task types; excluding unscored instance detection and endoscopy, our evaluation covers 17 datasets, seven modalities, and six scored task families: classification, multi-label classification, counting, detection, regression, and report generation. A single shared adaptation may suffer from interference between these tasks, whereas a separate fine-tuned VLM per task would substantially increase training, storage, and deployment costs.

Related Work. General-domain VLMs such as BLIP-2 [11] and Qwen2.5-VL [2] align visual and language representations through lightweight bridging modules or dynamic-resolution processing, while medical systems including LLaVA-Med [10], HuatuoGPT-Vision [4], and MedGemma [16] add medical image–text training or domain-specific visual representations. Yet parsing outputs differ in visual evidence, supervision, and decoding constraints. Previous challenge solutions to FLARE 2025 Task 5 [12] use different forms of specialization: Qwen-VLConnector [13] fuses multi-level visual features, ME-VLIP [17] routes questions among task-specific QLoRA adapters and a multi-task fallback, and Sun et al. [19] use a medically pretrained vision encoder with a shared LoRA-adapted language model. These approaches show that broad task coverage is possible but leave open how task-specific adaptation, prompting, and visual resolution should be allocated within one shared, deployable backbone. LoRA [8], QLoRA [5], and LLaMA-Factory [23] make such modular specialization practical.

Contributions. Our system uses Qwen3.5-9B [14] with a question-only GLiClass router [18] (DeBERTa-V3 [7]) selecting one of six specialist adapters or a general fallback on one shared 4-bit backbone. Our contributions are:

- We develop and validate a parameter-efficient system for six medical image-parsing tasks using one quantized Qwen3.5-9B backbone and lightweight task-specific LoRA adapters.
- We compare the adapted system with its zero-shot Qwen3.5-9B backbone on an internal holdout and public validation data.
- We compare specialist and general adapters, measure task-dependent resolution effects, and analyze errors hidden by aggregate metrics.

2 Method

We use one quantized Qwen3.5-9B backbone with task-specific LoRA adapters. A question-only router selects the adapter, image budget, and prompt for each case (Fig. 1).

2.1 Data preparation and image handling

We audited the official FLARE 2026 Task 3 2D data [6] and excluded 73 records that failed task-specific validity checks: 47 endoscopy records with invalid labels, 17 BUSI detection records with inverted or zero-area boxes, seven periapical records with contradictory labels, one duplicated periapical record, and one Boneresorption record with an extreme target value of 2,085 (dataset median: 28). We also excluded all 1,248 instance-detection samples because this task was not scored, and omitted endoscopy from adapter training because of its few usable examples, long multi-frame inputs, and noisy labels.

Following ME-VLIP [17], we expanded each multi-image training example into separate single-image examples, repeating the question and answer for each image. Questions sharing an image remained in the same partition. In earlier Qwen3-VL-8B experiments, grouping three neojaundice images reduced balanced accuracy in two complementary train–validation folds; this motivated, but does not validate, the policy for Qwen3.5-9B. At inference, the system accepts up to four images and selects four evenly spaced images from longer sequences.

Image preprocessing consists of RGB conversion and aspect-ratio-preserving resizing when an image exceeds its route's pixel-area cap. The cap is a maximum area, not a forced square shape. We used no additional image transformations, unlabeled examples, or partial labels.

2.2 Shared backbone and task routing

All routes use one Qwen3.5-9B backbone loaded with 4-bit NF4 quantization and bfloat16 computation. The pretrained backbone remains frozen. Six task-specific LoRA adapters and one general adapter, trained across all scored tasks, are loaded into the same model; switching tasks changes the active adapter without reloading the backbone.

We reuse the GLiClass classifier released with ME-VLIP [17], without further training. It uses a DeBERTa-v3-small encoder to predict a task label and confidence from the question text alone. The corresponding specialist is activated when the top confidence is at least 0.3 and the label has a configured route; otherwise, the general adapter is used. Supplied `TaskType` and `Modality` metadata do not influence routing.

2.3 Task-specific adaptation

We train each adapter by supervised fine-tuning, optimizing LoRA parameters with loss applied only to the answer tokens. Table 1 lists the deployed configurations. In the table, the full labeled pool comprises $A \cup B$, the internal holdout,

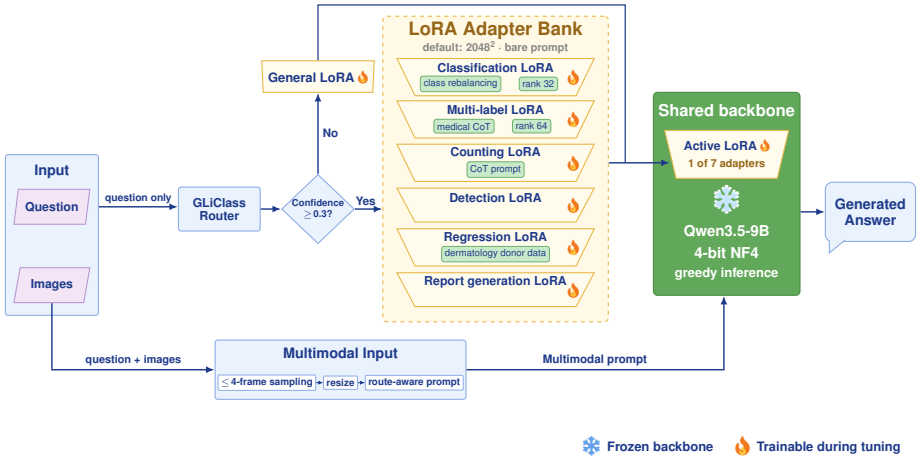


Fig. 1. Routed inference with a shared quantized backbone. The question selects a specialist or the general fallback, together with the image budget and prompt used for answer generation.

and `validation-public`; prompt variants are defined in Section 2.4. Shared optimization settings are given in Section 3.2.

Table 1. Per-route configuration of the proposed system. Pixel values denote maximum image-area budgets, not forced square resizing.

Route	Data scope	Pixels	LoRA r/α	Prompt	Output
Classification	Full labeled pool, oversampled	2048^2	32/64	Bare	Label string
Multi-label classification	Full labeled pool	2048^2	64/128	Medical	Label set
Counting	Full labeled pool	2048^2	8/16	Counting	Numeric string
Detection	Full labeled pool	2048^2	8/16	Bare	Nested boxes
Regression	A∪B + `bcn20000` donor mixture	2048^2	8/16	Bare	Numeric string
Report generation	A∪B	768^2	8/16	Bare	Free text
General adapter	A∪B, all scored tasks	768^2	8/16	Bare	Task-dependent

Preliminary Qwen3-VL-8B experiments informed the LoRA [8] ranks retained for Qwen3.5-9B: 32 for classification, 64 for multi-label classification, and 8 for the remaining adapters, with $\alpha = 2r$ throughout.

For classification, we use capped oversampling of rare dataset–answer groups. A group containing $n < 250$ training examples is expanded by duplication to $\min(6n, 250)$ examples; larger groups are unchanged. For regression, we com-

bine 5,265 examples from $A \cup B$ with 5,000 `bcn20000` dermatology-classification examples. The A/B partitions and internal holdout are defined in Section 3.1.

Adapters trained on $A \cup B$ provide the internal comparisons. For deployment, classification, multi-label classification, counting, and detection were subsequently refitted on the full labeled development pool. Regression, report generation, and the general fallback retain their $A \cup B$ adapters.

2.4 Prompting and inference

Each user message contains indexed image markers followed by the original question. Counting and multi-label classification also use the instructions below, placed before the question. They are inspired by chain-of-thought prompting [21], but training supervises only final answers. The remaining routes use the bare image–question prompt.

Task-specific prompts

Counting

System
You are a medical image analysis expert. Think carefully before answering.

Task instruction
Methodically count each object in the image (scan left-to-right, row by row). Write the final count as a single integer.

Multi-label classification

System
You are an expert radiologist and medical image analyst specialising in X-ray, CT, MRI, ultrasound, pathology, endoscopy, and microscopy. Think carefully and systematically before answering.

Task instruction
For each listed finding, assess whether it is clearly visible in the image. List only confirmed findings separated by commas.

Generation uses greedy decoding (`do_sample=False`), at most 512 new tokens, fixed end-of-sequence and padding identifiers, and Qwen3.5 thinking mode disabled. Post-processing removes any thinking tags, retains the final line for classification, normalizes multi-label delimiters, extracts detection boxes and numeric answers, and retains report text. If no numeric value is found, the numeric parser returns zero. We apply no calibration, re-querying, or manual answer correction.

The container packages the backbone, adapters, tokenizer, and router for fully offline execution. Its runtime, size, and repeated-run checks are reported in Section 4.4.

3 Experiments

3.1 Dataset and evaluation measures

Dataset. The cleaned source pool for FLARE 2026 Task 3 [6] contained 51,154 question–answer pairs from the organizer training and public-validation data. After the exclusions described in Section 2.1, our evaluation covers 17 datasets, seven imaging modalities, and six scored task families.

Following recommended cross-validation practice for medical-imaging AI [3], we partitioned the organizer training data by image group, stratified by dataset and task combination, into an approximately 10% internal holdout and two 45% subsets, A and B (Fig. 2). Smaller subsets $A' \subset A$ and $B' \subset B$, each containing 30% of its parent partition, supported inexpensive preliminary screening before selected configurations were evaluated on the full A/B partitions. Full $A \to B$ and $B \to A$ experiments were run where available. These partitions were fixed before expanding multi-image examples into single-image training rows.

Adapted models in the internal and public-validation comparisons are trained on $A \cup B$; public-validation images overlapping the organizer training data were excluded. Although internal-holdout images were excluded from these adapters' training, holdout results informed some development choices. We therefore treat the holdout results as development comparisons rather than independent estimates of generalization [20]. Selected deployment routes were later refitted with the holdout and public-validation data (Table 1); scores on these previously seen cases are not reported as held-out performance.

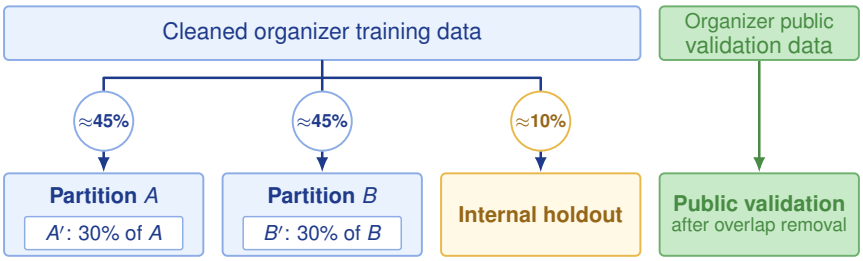


Fig. 2. Image-grouped development partitions and configuration-screening subsets. Percentages are approximate; A' and B' each contain 30% of their parent partition. Public-validation images overlapping the organizer training data were removed.

Evaluation. We use balanced accuracy (mean class recall) for classification and micro-averaged F1 for multi-label classification. Detection uses F1 with box matching at intersection-over-union > 0.5. Counting and regression use mean absolute error (MAE), and report generation uses the language-model-based CRIMSON metric [1].

We evaluate the internal and public splits separately, computing each metric per dataset and reporting its unweighted mean within each task. Counting

and report generation each contain one evaluated dataset. Excluding endoscopy, internal/public classification covers ten/six datasets, multi-label classification two/one, and detection three/two. Regression has two internal datasets and no public cases. Organizer hidden-validation scores (Section 4.1) instead pool cases within each task. Differences in dataset composition and aggregation prevent direct comparison with our dataset-macro results.

Statistical uncertainty. Paired 95% confidence intervals in the configuration ablations use 10,000 bootstrap resamples of matched question–answer cases, sampled with replacement within each dataset using the same sampled indices for both configurations. For the general-versus-specialist regression comparison, interval bounds are the 2.5th and 97.5th percentiles of the resampled MAE differences. This case-level resampling treats question–answer cases as independent; the intervals do not account for dependence between questions sharing an image or variation across training seeds.

Comparison protocols. We compare $A \cup B$ adapters with two zero-shot baselines on the same splits. The first uses the deployed prompts, answer prefixes, extraction rules, and 512-token limit. The format-assisted baseline adds explicit output-format instructions, removes the answer prefix, changes answer extraction, and allows 1,024 tokens. Both retain the same 4-bit Qwen3.5-9B backbone, greedy decoding, disabled thinking mode, and task-specific image budgets (768^2 for reports, 2048^2 otherwise). Thus, their difference measures a combined inference-protocol change, not a prompt-only effect. These evaluations select configurations using known task assignments and do not measure router accuracy or end-to-end container performance. Report generation uses only the first baseline; comparisons retain reports scored successfully for both models (780 internal and 1,943 public cases).

3.2 Implementation details

Environment settings Table 2 summarizes the single-GPU development environments used with LLaMA-Factory.

Table 2. Development environment.

Component	Setting
System	Ubuntu 22.04.5 LTS (A5000 workstation)
CPU / RAM	32 cores / 256 GB (A5000 workstation)
GPUs	NVIDIA RTX A5000 (24 GB)
	NVIDIA RTX PRO 6000 Blackwell (96 GB)
Framework	PyTorch 2.11.0, CUDA 12.8
Training library	LLaMA-Factory 0.9.5
Dependencies	Transformers 5.16.1, PEFT 0.18.1
	bitsandbytes 0.50.2
Code	https://github.com/TahaSabir0/FLARE26-Hexapus

Training protocols We applied QLoRA to all linear layers of the language model while keeping the pretrained weights, vision encoder, and multimodal projector frozen. Supervised fine-tuning used next-token cross-entropy loss on answer tokens only. Table 3 lists the shared settings. Runs used one GPU and an effective batch size of 8: batch size 2 with four accumulation steps on Blackwell, or batch size 1 with eight steps on the A5000.

An earlier single-seed Qwen3-VL-8B classification screen favored three epochs over two or four (0.5964 versus 0.5697 and 0.5876 dataset-macro balanced accuracy on $A' \to B'$ at 768^2). We transferred this schedule to Qwen3.5-9B without treating it as optimal for the final backbone or every task.

Table 3. Shared training settings for the QLoRA adapters.

Parameter	Setting
Epochs	3
Effective batch size	8
Optimizer	AdamW
Initial learning rate	2×10^{-4}
Learning rate schedule	Linear decay, 3% warm-up
Maximum sequence length	8,192 tokens
Compute precision	BF16
Gradient checkpointing	Enabled
Random seed	1337

Each adapter was evaluated at its training resolution using the final checkpoint, without early stopping, extra augmentation, or patch sampling. All reported Qwen3.5-9B experiments use one random seed; Section 3.1 describes configuration selection.

4 Results and discussion

4.1 Results on the validation sets

Internal and public validation.

Figure 3 compares the route-matched $A \cup B$ adapters with the two zero-shot protocols. These adapters exclude internal holdout and public-validation images from training; they are proxies for deployment routes subsequently refitted on the full labeled pool. Table 4 gives the internal per-dataset results.

Output conventions materially affect zero-shot performance. Format assistance raises classification balanced accuracy from 0.0035 to 0.3982 internally and from 0.0055 to 0.4229 publicly; adaptation then reaches 0.7791 and 0.7418. The same protocol change does not help multi-label classification or counting, and near-zero detection scores remain sensitive to coordinate conventions and answer extraction rather than measuring localization alone.

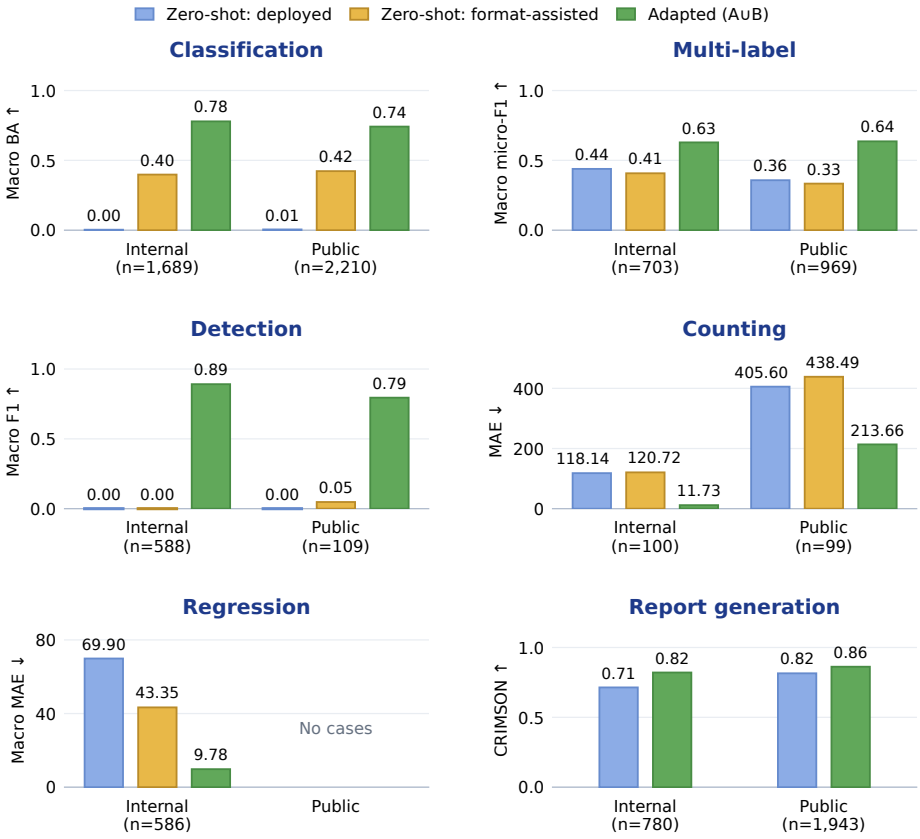


Fig. 3. Zero-shot and adapted Qwen3.5-9B on internal and public-validation data, using the protocols in Section 3.1. BA denotes balanced accuracy; multi-label F1 is micro-averaged within each dataset. Task scores are dataset macros, except single-dataset counting and report generation. Report means use the same successfully scored cases for both models; no format-assisted report run was performed. Public validation has no regression cases. Arrows indicate the better direction.

Adaptation improves every evaluated task over the available zero-shot protocols, although large errors remain. Relative to the deployed-settings baseline, counting MAE falls from 118.14 to 11.73 internally and from 405.60 to 213.66 publicly, where dense cell images remain difficult. On jointly scored reports, CRIMSON rises from 0.7141 to 0.8208 internally and from 0.8151 to 0.8622 publicly.

Organizer hidden validation. The hidden-validation submission was scored on 1,703 cases spanning classification, multi-label classification, and regression (Table 5). The official overall score was 0.4825. These metrics are computed over each task's full scoring surface; in particular, the challenge's sample-averaged

Table 4. Per-dataset results of the route-matched Qwen3.5-9B adapters on the internal holdout. The report score uses all 781 adapted-model scores; Fig. 3 uses the 780 reports jointly scored with zero-shot.

Task	Dataset	Metric	n	Resolution	Qwen3.5-9B
Classification	CMMD	BA	357	2048^2	0.5970
	BCN20000	BA	286	2048^2	0.6900
	Dental	BA	420	2048^2	0.6700
	Neojaundice	BA	60	2048^2	0.8100
	Bone marrow	BA	72	2048^2	0.8660
	BUS-UCLM	BA	52	2048^2	0.9190
	Fundus	BA	138	2048^2	0.6870
	IUGC	BA	131	2048^2	0.9300
	Retino	BA	111	2048^2	0.7610
	BUSI	BA	62	2048^2	0.8610
Multi-label	ChestDR	F1	388	2048^2	0.5625
	Periapical	F1	315	2048^2	0.6947
Counting	NeurIPS22Cell	MAE	100	2048^2	11.7300
Detection	IUGC	F1	516	2048^2	0.9981
	BUSI-det	F1	50	2048^2	0.8317
	BUS-UCLM-det	F1	22	2048^2	0.8444
Regression	Boneresorption	MAE	70	2048^2	17.1600
	IUGC	MAE	516	2048^2	2.4100
Report generation	IU-XRay	CRIMSON	781	768^2	0.8204

multi-label F1 is distinct from the per-dataset micro-F1 used in our internal and public-validation comparisons.

Table 5. Official hidden-validation performance of the submitted Qwen3.5-9B system. Scores are pooled by task.

Task	Metric	n	Score
Classification	Balanced accuracy ↑	1,126	0.8496
Multi-label classification	Sample-averaged F1 ↑	477	0.5338
Regression	MAE ↓	100	14.5950

4.2 Configuration ablations

General versus task-specific adapters. Task-specific configurations produced uneven benefits over the shared general adapter (Table 6(a)). All rows compare matched 768^2 input. Specialists score higher on four of six tasks, including the largest scored slice, classification, while the multi-label and report-generation comparisons show no clear difference.

On Boneresorption, the specialist reduces MAE by 3.93 (paired 95% confidence interval $[0.83, 7.06]$) and produces 12 exact-zero answers versus 29 for the general adapter. This supports the specialist recipe on that dataset, but does not measure router accuracy.

Table 6. Configuration ablations on the internal holdout. (a) General adapter versus task specialists at matched 768^2 input. (b) Resolution comparisons for task specialists. Regression is the mean MAE over two datasets.

(a) General versus task-specific adapters			
Task	Metric	General	Specialist
Classification	BA ↑	0.7348	**0.7695**
Multi-label classification	F1 ↑	**0.6157**	0.6130
Counting	MAE ↓	14.3400	**13.4500**
Detection	F1 ↑	0.8876	**0.9194**
Regression	MAE ↓	12.1600	**10.1800**
Report generation	CRIMSON ↑	**0.8307**	0.8204
(b) Image-resolution budget			
Task	Metric	768^2	2048^2
Classification	BA ↑	0.7695	**0.7791**
Multi-label classification	F1 ↑	0.6130	**0.6286**
Detection	F1 ↑	**0.9194**	0.8914
Regression	MAE ↓	10.1800	**9.7800**
Counting	MAE ↓	13.4500	**11.7300**

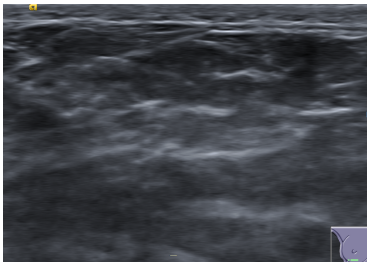
Image resolution. Raising the image budget from 768^2 to 2048^2 has mixed effects (Table 6(b)). Classification BA increases by 0.0096, multi-label F1 increases by 0.0156, counting MAE decreases by 13%, and regression macro MAE decreases by 0.40, while detection F1 decreases by 0.0280. The paired 95% interval for the multi-label change is $[-0.004, +0.035]$, and both per-dataset regression intervals include zero. Runs also used different GPUs and micro-batch shapes at the same effective batch size, so the changes cannot be attributed to image detail alone. Although the 768^2 $A \cup B$ detection adapter scored 0.9194, it is not the deployed route: the full-data detection adapter was trained at 2048^2. Table 4 therefore reports the route-matched $A \cup B$ proxy at 2048^2 (0.8914 F1). Regression likewise remains at 2048^2, while report generation uses 768^2.

4.3 Strengths and remaining failure modes

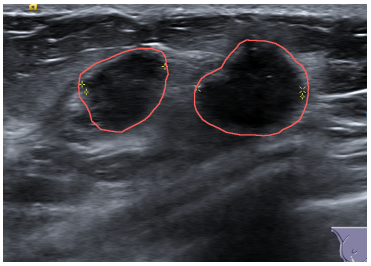
An output-format audit of internal adapted-model predictions found valid nested coordinate lists in all 588 detection responses, valid integers in all counting responses, and no empty answers or leaked reasoning tags. The remaining errors mainly reflect visual recognition and numerical estimation rather than syntax.

Figure 4 illustrates representative successes and task-specific failures across categorical, numerical, and spatial outputs. All displayed outputs are syntactically valid; the remaining errors arise from visual category confusion, a degenerate numerical estimate, or incomplete localization.

Report generation remains the main semantic limitation. Among 223 cases with abnormal reference reports, 137 (61.4%) receive normal-only predictions. Only 16.4% of predictions are unique, and the ten most frequent outputs account for 60.6%, indicating heavy reuse of report templates. The resulting omissions are not apparent from the aggregate CRIMSON score of 0.8204.



(a) Classification, BUS-UCLM.
Question: What does the contour-defined zone in the breast ultrasound scan signify? Options: A) Benign lesion present; B) Malignant lesion present; C) No lesion, only normal tissue.
Ground truth: C. **Prediction:** C.



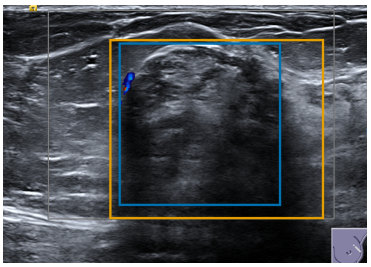
(b) Classification, BUS-UCLM.
Question: What does the contour-defined zone in the breast ultrasound scan signify? Options: A) Benign lesion present; B) Malignant lesion present; C) No lesion, only normal tissue.
Ground truth: B. **Prediction:** A.



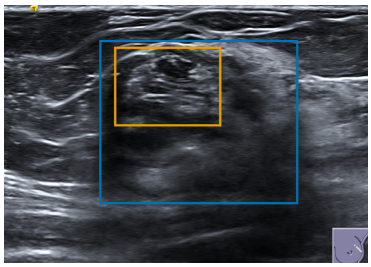
(c) Regression, Boneresorption.
Question: Assess the rate of alveolar bone resorption (ABR%) from this orthopantomogram. **Ground truth:** 24.2.
Prediction: 25.0. **Absolute error:** 0.8.



(d) Regression, Boneresorption.
Question: Estimate the alveolar bone resorption percentage (ABR%) based on the radiographic findings in this panoramic image.
Ground truth: 31.1. **Prediction:** 0.0.
Absolute error: 31.1.



(e) Detection, BUS-UCLM-det. IoU: 0.68.
Question: Report the spatial coordinates of lesions in the ultrasound scan as $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$, structuring multiple lesions as a nested list.
Ground truth: `[[274,90,651,469]]`.
Prediction: `[[252,82,752,500]]`.



(f) Detection, BUS-UCLM-det. IoU: 0.25.
Question: Pinpoint lesion regions in the ultrasound image with $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ coordinates, nesting lists for multiple findings.
Ground truth: `[[226,84,689,465]]`.
Prediction: `[[261,100,508,282]]`.

Fig. 4. Representative successes or low-error predictions (left) and failures (right). Classification and detection use non-overlapping `validation-public` cases; regression uses the internal holdout because public validation has no regression cases. Blue boxes show ground truth and orange boxes show predictions. All examples use task-specific $A \cup B$ adapters.

Table 7. Final container verification on 32 cases covering all six tasks.

Measure	Observed	Limit or check
Mean runtime (incl. model loading)	3.1 s/case	$\leq$ 6.57 s/case
Container image size	30.1 GB	$\leq$ 35 GB
Repeated-run identity	32/32 cases	Identical outputs

4.4 Deployment efficiency

The final container meets the runtime and size limits and produces identical outputs across two runs (Table 7).

4.5 Results on the Final Testing Set

This is a placeholder. We will present the testing results during MICCAI.

4.6 Limitations and future work

The organizer hidden-validation score covers classification, multi-label classification, and regression only, so evidence for counting, detection, and report generation is limited to the internal and public-validation sets, which differ in composition and are not directly comparable. Our Qwen3.5-9B experiments also use a single training seed. The 2048^2 production-adapter runs peaked at approximately 40–44 GB of GPU memory, requiring a 48 GB-class or larger GPU, whereas the lower-resolution runs fit on our 24 GB workstation GPUs; the resolution comparisons also differed in GPU class and micro-batch shape at the same effective batch size. Future work should confirm these task-dependent effects across seeds, quantify errors introduced by the question-only router, and investigate selective high-resolution processing or adaptive tiling under matched compute budgets. A promising extension would combine our adapter routing with the multi-level visual feature fusion of QwenVLConnector [13], letting the router also select a specialized visual pathway for fine spatial detail.

5 Conclusion

We demonstrate that a single quantized VLM can support six heterogeneous medical image-parsing tasks without a separate backbone for each one. The submitted 30.1 GB system averaged 3.1 seconds per case and achieved an overall score of 0.4825 on organizer hidden validation across classification, multi-label classification, and regression. Our controlled comparisons show no universal adaptation recipe: specialist adapters improved selected structured tasks, higher resolution substantially benefited dense counting but produced smaller or inconsistent gains elsewhere, and the general adapter remained competitive for some routes. Report generation exposed a different limitation: structurally valid outputs could still omit clinically relevant findings, with 61.4% of abnormal-reference cases receiving normal-only predictions. A shared backbone should

therefore unify deployment, not force uniform training or inference; future generalist medical VLMs should allocate adapter capacity and visual detail by task demand and be evaluated per dataset and failure mode, not aggregate scores.

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Disclosure of Interests

The authors declare no competing interests.

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Table 8. Checklist Table. Please fill out this checklist table in the answer column.

Requirements	Answer
A meaningful title	Yes
The number of authors (≤6)	6
Author affiliations and ORCID	Yes
Corresponding author email is presented	Yes
Validation scores are presented in the abstract	Yes
Introduction includes at least three parts: background, related work, and motivation	Yes
A pipeline/network figure is provided	Figure 1
Pre-processing	Section 2.1
Strategies to use the partial label	N/A
Strategies to use the unlabeled images.	N/A
Strategies to improve model inference	Section 2.4
Post-processing	Section 2.4
The dataset and evaluation metric section are presented	Section 3.1
Environment setting table is provided	Table 2
Training protocol table is provided	Table 3
Ablation study	Table 6
Efficiency evaluation results are provided	Table 7
Visualized segmentation example is provided	N/A (no segmentation task)
Limitation and future work are presented	Yes
Reference format is consistent.	Yes